

Kason Stubbart

Provo, UT • 602-663-3768 • kgstubbart@gmail.com • [linkedin.com/in/kason-stubbart-937015249](https://www.linkedin.com/in/kason-stubbart-937015249) • <https://github.com/kgstubbart>

Computer Science student at Brigham Young University committed to creating software that delivers practical results. Built a fully functional chess server, the Library Ace web platform, and analytical projects, including research on networks of power and poverty, showcasing a blend of technical skill and ingenuity. Skilled in full-stack development, software architecture, and developing functional, user-focused systems. Eager to leverage creativity, communication, and coding expertise to collaborate with your team in designing and implementing high-quality software.

SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, TypeScript

Software & Web Development: React, Node.js, REST APIs, WebSocket, HTML5, CSS, UI Design

Databases & Cloud: MySQL, AWS EC2, AWS Lambda, AWS SQS, Cloud Deployment, AWS DynamoDB

Tools & Version Control: Visual Studio Code, IntelliJ, PyCharm, Git, GitHub, GitBash, PowerShell

WORK EXPERIENCE

Computer Science Research Assistant

08/2024 - Present

Human-Centered Machine Intelligence Lab at BYU

Advancing research on networks of power and poverty by designing and implementing a visualization pipeline to model resource and influence flows on complex networks, with the goal of informing interventions to reduce poverty.

- Designed and validated a round-based token-exchange simulation and test suite, enabling reproducible experiments across 112 distinct network configurations to evaluate intervention strategies.
- Developed advanced visualization tools that enhanced experiment throughput by 55%, reducing analysis time from hours to minutes.
- Presented findings at a research conference attended by over 600 professionals.

Instructional Design Assistant Lead

04/2024 - 08/2024

BYU Independent Study

Led creation of educational materials, guiding multiple teams in developing course content and learning activities. I ensured the curriculum met goals, assisted with project management, and integrated technology to improve project efficiency.

- Led and contributed to 3–4 projects at a time, writing, editing, and organizing over 100 pages of material.
- Implemented a new AI tool, resulting in a 39% increase in efficiency and cost savings, driving improved project delivery timelines.
- Provided leadership to 30 employees through open communication channels and training presentations.

Instructional Design Assistant

02/2023 - 04/2024

BYU Independent Study

- Increased project cross-compatibility by 32% while completing several projects using popular Learning Management Systems.
- Wrote and edited HTML interfaces for several course facets, including videos, layout, and accessibility.
- Applied design styles to provide a focused UI for students to interact with in lesson activities.

PROJECTS

Exoplanet Atmosphere Analyzer

Developing a Python-based tool for analyzing exoplanet transmission spectra to characterize atmospheric composition. The project aims to identify common atmospheric gases and estimate planetary temperature from spectroscopic data.

Library Ace

Library Ace is an AWS-hosted web application that allows users to favorite a book, contributing to a community-driven ranking of most loved titles. Users can browse book details and discover popular reads based on collective favorites. Built using React, JavaScript, CSS, HTML, MySQL, and the Google Books API.

Networks of Power and Poverty

Human-Centered Machine Intelligence Lab at BYU

Developed a simulation and visualization tool to analyze how resources flow through social networks, supporting research on poverty and inequality. Implemented token exchange mechanisms across multiple network topologies with interactive visualizations and customizable parameters, applying Python, algorithm design, and data visualization skills.

EDUCATION

Bachelor of Science in Computer Science

Brigham Young University

08/2022 - 04/2027

Minor in Astronomy

Brigham Young University

08/2022 - 04/2027

High School Diploma

Perry High School

07/2016 - 05/2020

Magna Cum Laude

AWARDS & SCHOLARSHIPS

BYU Management Society Scholarship

04/2024

BYU Management Society

National Honor Society

04/2020

VOLUNTEERING & LEADERSHIP

The Church of Jesus Christ of Latter-Day Saints

07/2020 - 06/2022

Church Missionary • Orem, UT

BYU Rocketry

BYU Astronomy

Ukulele Club

President

Feed My Starving Children

Volunteer